DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli
Kanakapura Road, Bengaluru South Dt - 562112, Karnataka, India

SCHOOL OF ENGINEERING

Bachelor of Technology in
COMPUTER SCIENCE AND ENGINEERING

Major Project Phase-II Report

SMARTSHIELD: REAL-TIME URL-BASED PHISHING DETECTION USING MACHINE LEARNING

Batch : 145

By
SAVITA- ENG22CS0156.
SHASHIKALA- ENG22CS0163.
SNEHA- ENG22CS0173.
SNEHA M P- ENG22CS0176.

Under the supervision of Dr.
Sivananda Lahari Reddy Associate Professor
Dept. of CSE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, SCHOOL OF ENGINEERING
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SCHOOL OF ENGINEERING

Department of Computer Science & Engineering
Devarakaggalahalli, Harohalli, Kanakapura Road, Bengaluru South Dt - 562112
Karnataka, India

CERTIFICATE

This is to certify that the Major Project Stage-I work titled **"SMARTSHIELD: REAL-TIME URLBASED PHISHING DETECTION USING MACHINE LEARNING"** is carried out by **Savita (ENG22CS0156), Shashikala (ENG22CS0163), Sneha (ENG22CS0173), Sneha M P (ENG22CS0176),** bonafide students seventh semester of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

Dr. Sivananda Lahari Reddy	**Dr. Girisha G S**	**Dr. Udaya Kumar Reddy K R**
Associate Professor Dept. of CS&E, School of Engineering Dayananda Sagar University	Chairman CSE School of Engineering Dayananda Sagar University	Dean School of Engineering Dayananda Sagar University
Date:	Date:	Date:

Name of the Examiner **Signature of Examiner**

DECLARATION

We, **Savita (ENG22CS0156), Shashikala (ENG22CS0163), Sneha (ENG22CS0173), Sneha M P (ENG22CS0176),** are students of seventh semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, hereby declare that the Major Project Pahse-I titled **"SmartShield: Real-Time URL-based Phishing Detection using Machine Learning"** has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026.**

Student **Signature**

Name1 : Savita

USN : ENG22CS0156

Name2 : Shashikala

USN : ENG22CS0163

Name3 : Sneha

USN : ENG22CS0173

Name4 : Sneha M P

USN : ENG22CS0176

Place : Bangalore

Date :

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
DL	Deep Learning
URL	Uniform Resource Locator
ML	Machine Learning
NLP	Natural Language Processing
RF	Random Forest
CL	Contrastive Learning
API	Application Programming Interface
EDA	Exploratory Data Analysis
TF-IDF	Term Frequency–Inverse Document Frequency
CNN	Convolutional Neural Network
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve

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ABSTRACT

Phishing attacks have become one of the most serious cybersecurity threats, where attackers create fraudulent websites to deceive users into revealing sensitive information such as usernames, passwords, and financial details. Traditional phishing detection techniques, such as blacklist-based and rule-based approaches, are ineffective against newly generated (zero-day) phishing websites and often introduce latency and privacy concerns.This project proposes SmartShield, a lightweight real-time phishing detection system that utilizes a two-layer detection approach combining URL-based analysis and HTML content analysis. The system employs efficient text processing techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) to convert textual data into numerical feature vectors, and cosine similarity to measure the similarity between suspicious webpages and known phishing patterns.

In the first layer, the system analyzes the structure of URLs to identify suspicious patterns such as abnormal length, special characters, and misleading keywords. In the second layer, the HTML content of the webpage is examined to detect hidden phishing indicators such as fake login forms and malicious scripts. The combination of these two layers significantly improves detection accuracy and reduces false positives.The system is implemented using a browser extension integrated with a lightweight backend server, enabling real-time phishing detection without relying on cloud-based services. When a user visits a webpage, the system captures the URL and content, processes it using TF-IDF and cosine similarity, and classifies it as safe, suspicious, or phishing.

Experimental results demonstrate that the proposed system achieves fast and efficient phishing detection with low computational overhead, making it suitable for real-time deployment in browser environments. The SmartShield system enhances user security by providing timely warnings and helping prevent potential data theft and cyber-attacks.

CHAPTER 1 INTRODUCTION

The rapid growth of internet usage and online services has significantly increased the risk of cyber threats, among which phishing attacks are one of the most prevalent and dangerous. Phishing is a form of cyber-attack where attackers create fraudulent websites that closely resemble legitimate ones to deceive users into revealing sensitive information such as usernames, passwords, banking details, and personal data. As these attacks continue to evolve in sophistication, it becomes increasingly difficult for users to distinguish between genuine and malicious websites, leading to serious financial loss and privacy breaches.

Traditional phishing detection methods primarily rely on blacklist-based approaches, rule-based filtering, or cloud-based reputation systems. While these techniques are effective in detecting known phishing websites, they fail to identify newly generated or zero-day attacks. Additionally, cloud-based solutions introduce latency, depend on continuous internet connectivity, and may compromise user privacy by transmitting browsing data to external servers. These limitations highlight the need for a lightweight, real-time, and privacy-preserving phishing detection system that can operate efficiently without relying on external services.

To address these challenges, this project proposes SmartShield, a lightweight real-time phishing detection system that uses a two-layer detection approach combining URL-based analysis and HTML content analysis. The system employs efficient text processing techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) to convert textual data into numerical feature vectors, and cosine similarity to measure the similarity between suspicious webpages and known phishing patterns. Implemented as a browser extension integrated with a lightweight backend server, the system captures URLs and webpage content in real time, processes them locally, and classifies webpages as safe, suspicious, or phishing. This approach ensures faster response time, low computational overhead, improved detection accuracy, and enhanced user privacy, making it suitable for real-world deployment.

1.1. FIGURES AND TABLES

Sl. No.	URL Type	Source	Purpose in System
1	Legitimate URLs	Kaggle	Model training and testing
2	Phishing URLs	Kaggle	Model training and testing
3	Unlabeled URLs	PhishTank	Contrastive pretraining

Table 1.1 : Categories of URLs Used in Dataset

1.2. SCOPE

The scope of the SmartShield system is to design and implement a lightweight, real-time phishing detection framework that identifies malicious websites using both URL structure and webpage HTML content. The system analyzes URLs for suspicious patterns and processes webpage content using efficient text processing techniques such as TF-IDF and cosine similarity to classify websites as safe, suspicious, or phishing.

The system is implemented as a browser extension integrated with a local backend server, enabling real-time detection with low latency and improved user privacy. However, the scope is limited to web-based phishing detection in desktop browsers and does not include email phishing detection, mobile platforms, or cloud-based deployment.

CHAPTER 2 PROBLEM DEFINITION

Phishing attacks have become one of the most significant cybersecurity threats, where attackers create fraudulent websites that closely resemble legitimate ones to steal sensitive user information such as login credentials, banking details, and personal data. These phishing websites are continuously evolving, making it difficult for users to identify them and for traditional security systems to detect them effectively.

Existing phishing detection methods mainly rely on blacklist-based approaches, rule-based filtering, or cloud-based reputation systems. These approaches have several limitations. Blacklists are ineffective against newly generated (zero-day) phishing websites, rule-based systems require frequent updates and fail against obfuscated URLs, and cloud-based systems introduce latency, depend on internet connectivity, and may compromise user privacy.

Therefore, there is a need for a lightweight, real-time, and privacy-preserving phishing detection system that can accurately identify phishing websites without relying on external services. The proposed SmartShield system addresses this problem by using a two-layer detection approach that analyzes both URL structure and webpage HTML content. By applying efficient techniques such as TF-IDF and cosine similarity, the system aims to detect phishing websites quickly and accurately, including previously unseen attacks, while maintaining low computational overhead.

CHAPTER 3 LITERATURE REVIEW

Phishing attacks remain one of the most prevalent cybersecurity threats, exploiting deceptive URLs to trick users into revealing sensitive information. Early research in phishing detection primarily relied on blacklist-based mechanisms, which compare URLs against known malicious databases. Although effective for known threats, blacklist approaches fail to detect zero-day phishing URLs and require constant updates, making them insufficient for real-time protection [1][2].

To overcome these limitations, researchers introduced machine learning-based phishing detection techniques that analyze URL characteristics instead of relying solely on blacklists. Lexical features such as URL length, presence of special characters, number of subdomains, entropy, and suspicious keywords have shown strong predictive power in distinguishing phishing URLs from legitimate ones [3][4]. Studies using classical machine learning algorithms like Support Vector Machines (SVM), Naïve Bayes (NB), Decision Trees (DT), and Random Forests (RF) demonstrated high accuracy, often exceeding 90%, with Random Forest consistently outperforming other classifiers due to its robustness against overfitting and ability to handle high- dimensional data [5][6].

Further advancements incorporated Natural Language Processing (NLP) techniques to extract meaningful patterns from URL text. Approaches using TF-IDF vectorization combined with machine learning classifiers have achieved improved detection accuracy while maintaining low computational overhead, making them suitable for real-time systems [7][8]. Several works confirmed that purely URL-based models, without accessing webpage content, can still achieve strong performance, thereby preserving user privacy and reducing latency [9].

Recent studies explored hybrid and ensemble approaches that combine multiple feature types, such as lexical, statistical, and domain-based features. These hybrid models demonstrated better generalization and improved zero-day phishing detection compared to single-feature approaches [10][11]. Additionally, deep learning techniques, including Capsule Networks and transformer- based models like BERT, have been applied to phishing URL detection, showing superior

performance in capturing semantic relationships within URLs. However, these models often require higher computational resources, making deployment in lightweight environments such as browser extensions more challenging [12][13].

Despite the progress, most existing solutions depend on cloud-based analysis, third-party APIs, or centralized servers, raising privacy concerns and increasing response latency. Moreover, many systems focus on achieving high accuracy in offline evaluation but lack practical real-time deployment mechanisms. These gaps highlight the need for a lightweight, privacy-preserving, and real-time phishing detection system that operates locally while maintaining high detection accuracy. The proposed system addresses these challenges by combining efficient feature extraction with machine learning models optimized for real-time URL classification [14][15].

CHAPTER 4 PROJECT DESCRIPTION

Phishing websites are becoming increasingly advanced, making it difficult for traditional detection systems to identify them effectively. The objective of this project is to develop SmartShield, a lightweight and real-time phishing detection system capable of detecting both known and zero-day phishing attacks. The system is designed to be efficient, privacy-preserving, and suitable for deployment in a browser-based environment.

The SmartShield system uses a two-layer detection approach. In the first layer, the system analyzes the structure of URLs to identify suspicious patterns such as unusual length, special characters, and misleading keywords. In the second layer, the system examines the HTML content of the webpage to detect hidden phishing indicators such as fake login forms, suspicious scripts, and deceptive elements. This combined approach improves detection accuracy and reduces false predictions..

4.1 PROPOSED DESIGN

The proposed SmartShield system is designed to provide real-time phishing detection using a combination of URL analysis and HTML content analysis. The system follows a hybrid architecture consisting of a browser extension and a local backend server.

When a user visits a webpage, the browser extension captures the URL and extracts the HTML content. This information is sent to the backend server for processing. The backend applies Term Frequency–Inverse Document Frequency (TF-IDF) to convert textual data into numerical feature vectors. Cosine similarity is then used to measure the similarity between the given webpage and known phishing patterns.

Based on the similarity score, the system classifies the webpage as safe, suspicious, or phishing. If a phishing website is detected, the browser extension displays a warning message to alert the user before interaction. This approach ensures fast detection, low computational overhead, and improved user privacy as all processing is performed locally without relying on external cloud services.

4.2 ASSUMPTIONS AND DEPENDENCIES

Assumptions :

- It is assumed that the URLs provided to the system follow standard formatting rules.
- The system assumes that phishing websites exhibit identifiable patterns in both URL structure and HTML content.
- The dataset used for comparison contains representative phishing patterns for similarity analysis.
- The browser environment supports extension-based URL monitoring and content extraction.
- The system assumes sufficient local computational resources for real-time processing.

Dependencies :

- The system depends on browser APIs to capture URLs and extract HTML content.
- It requires text processing libraries for implementing TF-IDF vectorization.
- Cosine similarity computation depends on numerical feature representations generated from textual data.
- The backend server (e.g., Flask) is required for processing and classification.
- The performance of the system depends on the quality and diversity of the dataset used for similarity comparison.

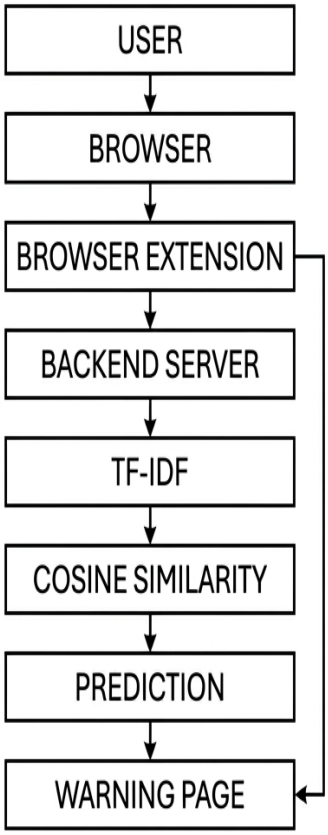

Figure 4.1: SmartShield Workflow Diagram

CHAPTER 5 REQUIREMENTS

5.1 FUNCTIONAL REQUIREMENTS

- The system shall capture URLs automatically when a user visits a webpage.
- The system shall extract HTML content from the webpage.
- The system shall send the captured URL and content to the backend server for processing.
- The system shall apply TF-IDF to convert textual data into numerical feature vectors.
- The system shall compute cosine similarity to compare webpage data with known phishing patterns.
- The system shall classify webpages as safe, suspicious, or phishing.
- The system shall display a warning message if a phishing website is detected.
- The system shall allow the user to either proceed or go back after a warning.

5.2 NON-FUNCTIONAL REQUIREMENTS

- Performance: The system should provide real-time detection with minimal delay.
- Accuracy: The system should achieve high detection accuracy with low false positives.
- Privacy: The system should not send user data to external servers.
- Security: The system should ensure safe handling of user browsing data.
- Usability: The system interface should be simple and user-friendly.
- Reliability: The system should function consistently during continuous browsing.
- Maintainability: The system should allow easy updates and improvements.

5.3 SOFTWARE REQUIREMENTS

- Programming Languages: Python, JavaScript
- Backend Framework: Flask
- Libraries: Scikit-learn (TF-IDF), NumPy
- Database: SQLite (for feedback storage)
- Development Tools: VS Code / PyCharm
- Browser API: Chrome Extension APIs (Manifest V3)

CHAPTER 6 METHADOLOGY

The methodology of the SmartShield system consists of a sequence of steps designed to detect phishing websites in real time using a combination of URL analysis and HTML content analysis. The system follows a two-layer detection approach and uses lightweight text processing techniques such as TF-IDF and cosine similarity to ensure efficient and accurate classification.

6.1 Data Collection And Preprocessing

The dataset used for the system includes both phishing and legitimate website data. Phishing data is collected from publicly available sources, while legitimate URLs are obtained from trusted domains. The collected data is cleaned by removing duplicates, invalid entries, and unnecessary characters. Text normalization techniques such as lowercasing and removal of special symbols are applied to ensure consistency before processing.

6.2 Feature Extraction

Feature extraction is performed on both URL and HTML content. The system identifies important patterns such as URL length, presence of suspicious keywords, number of special characters, and structural properties. Additionally, the textual data is converted into numerical feature vectors using TF-IDF (Term Frequency–Inverse Document Frequency), which assigns importance to words based on their frequency and relevance.

6.3 Similarity Measurement

After feature extraction, cosine similarity is used to measure the similarity between the processed webpage data and known phishing patterns. Cosine similarity calculates the angle between two feature vectors and determines how closely they are related. A higher similarity score indicates a higher probability that the webpage is phishing.

6.4 Classification

Based on the similarity score, the system classifies the webpage into one of three categories: safe, suspicious, or phishing. Threshold values are defined to determine the classification. If the similarity score exceeds a certain threshold, the webpage is marked as phishing and appropriate action is taken.

6.5 System Deployment

The SmartShield system is deployed using a hybrid architecture consisting of a browser extension and a local backend server. The browser extension captures the URL and extracts HTML content whenever a user visits a webpage. This data is sent to the backend server, where feature extraction, TF-IDF processing, and cosine similarity calculations are performed. The result is then returned to the extension, which displays a warning message if necessary.

6.6 System Workflow

The overall workflow of the system is as follows:

- User visits a webpage
- Browser extension captures the URL
- HTML content is extracted
- Data is sent to the backend server
- TF-IDF is applied to convert text into vectors
- Cosine similarity is computed
- Webpage is classified as safe, suspicious, or phishing
- Warning message is displayed if required

CHAPTER 7 EXPERIMENTATION

This chapter describes the implementation and testing of the SmartShield phishing detection system. The system is developed using a combination of a browser extension and a local backend server. The experimentation focuses on evaluating the system's ability to detect phishing websites in real time using URL and HTML content analysis.

7.1 Implementation Setup

The SmartShield system is implemented using Python for backend processing and JavaScript for the browser extension. The backend server is developed using the Flask framework, which handles data processing and communication with the extension. TF-IDF vectorization and cosine similarity calculations are performed using libraries such as Scikit-learn and NumPy.

The browser extension is developed using Chrome Extension APIs, which capture URLs and extract HTML content when a user visits a webpage. The captured data is sent to the backend server, where it is processed and classified. The system runs on a local machine, ensuring faster response time and improved privacy.

7.2 Testing Approach

The system is tested using a set of phishing and legitimate websites. Different scenarios are considered, including known phishing websites, newly generated suspicious URLs, and legitimate websites. The testing process involves verifying whether the system correctly classifies each webpage based on its content and structure.

The testing also evaluates how quickly the system responds when a user visits a webpage and whether the warning messages are displayed correctly. Multiple test cases are executed to ensure the reliability and consistency of the system.

7.3 Performance Evaluation

The performance of the SmartShield system is evaluated using standard metrics such as accuracy, precision, recall, and response time. Accuracy measures the overall correctness of the system, while precision and recall evaluate the system's ability to correctly identify phishing websites without generating false alerts.

The system demonstrates efficient performance with fast response times due to the use of lightweight techniques such as TF-IDF and cosine similarity. The results indicate that the system is capable of detecting phishing websites effectively while maintaining low computational overhead, making it suitable for real-time deployment.

7.4 Test Cases

Some sample test cases used during experimentation are shown below:

Test Case	Input Type	Expected Output	Actual Output	Result
1	Phishing URL	Phishing	Phishing	Pass
2	Legitimate URL	Safe	Safe	Pass
3	Suspicious URL	Suspicious	Suspicious	Pass
4	Phishing HTML	Phishing	Phishing	Pass
5	Normal Website	Safe	Safe	Pass

CHAPTER 8 RESULTS

- Displays real-time URL classification as SAFE or PHISHING with confidence score using TF-IDF and cosine similarity.

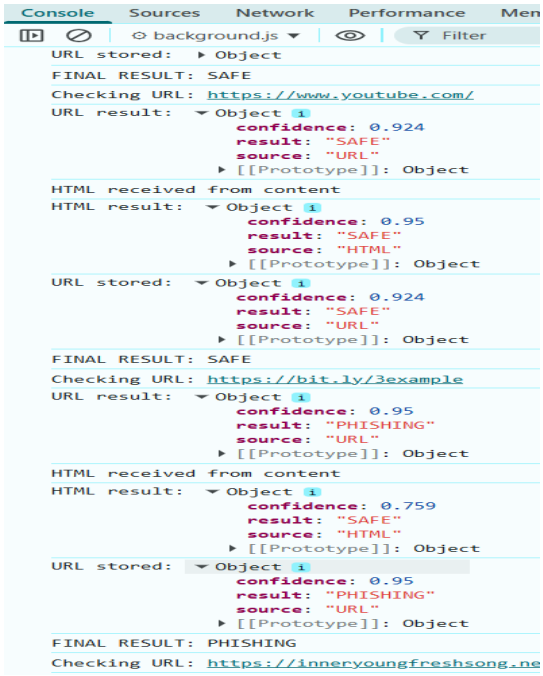

Figure 8.1: URL Prediction Results in Browser Console

- Shows server-side processing of URL and HTML content with prediction results and confidence values.

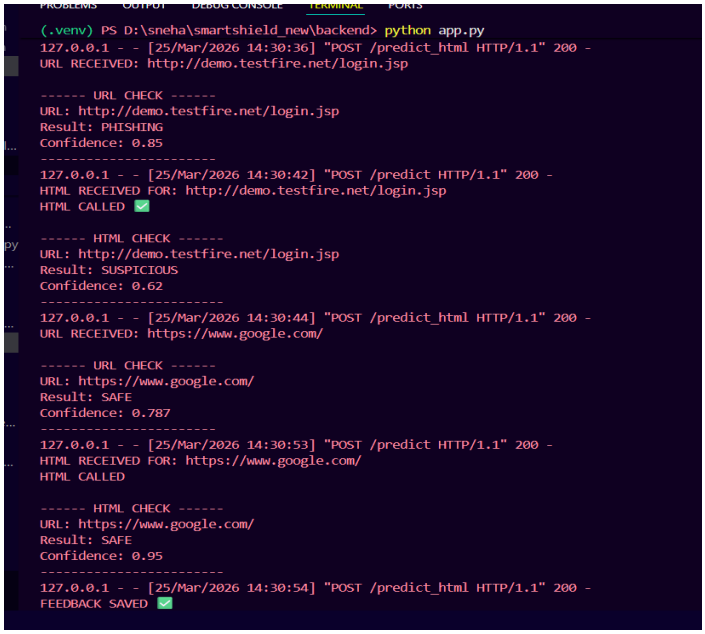

Figure 8.2: Backend Terminal Showing URL and HTML Analysis

- Displays stored phishing detection results along with user validation feedback for performance tracking.

28	https://slowgoodbeautifulmagic.neverssl.com/online/	PHISHING	CORRECT	2026-03-25 12:59:12
29	http://demo.testfire.net/login.jsp	PHISHING	CORRECT	2026-03-25 13:01:45
30	http://demo.testfire.net/login.jsp	PHISHING	CORRECT	2026-03-25 13:02:34

Figure 8.3: Stored Prediction Results with User Feedback

- Shows the warning interface triggered when a phishing website is detected, allowing user action and feedback.

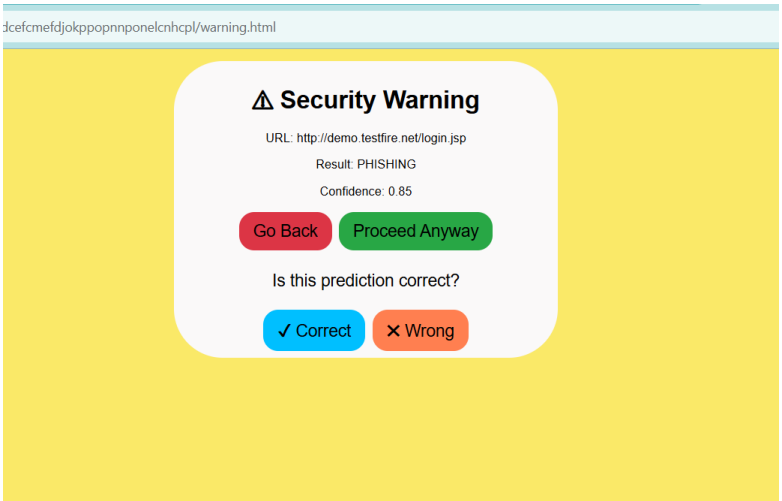

Figure 8.4: Phishing Warning Page Displayed to User

- Shows successful installation of the SmartShield extension used for real-time phishing detection in the browser.

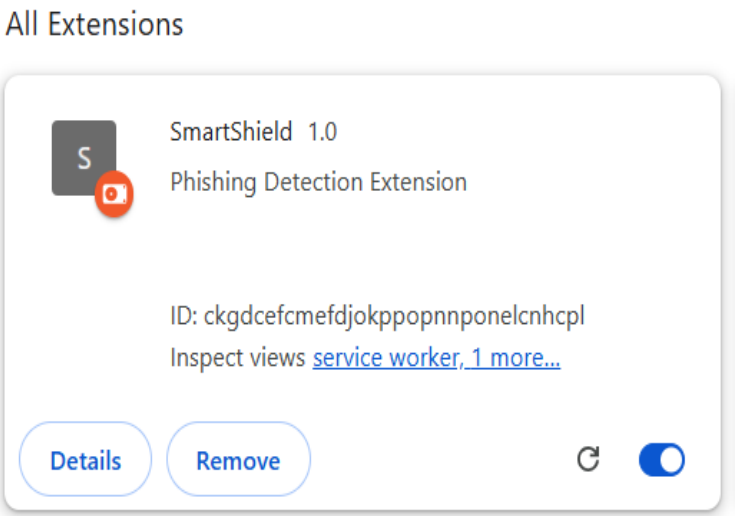

Figure 8.5: Smartshield Chrome Extension Installation

CHAPTER 9 DELIVERABLES

The SmartShield project delivers a functional and efficient phishing detection system capable of identifying malicious websites in real time. The system includes both frontend and backend components, working together to provide accurate and fast detection.

The key deliverables of the project are as follows:

- A browser extension that captures URLs and extracts webpage HTML content in real time
- A backend server developed using Flask for processing and classification
- Implementation of TF-IDF for feature extraction from URL and HTML content
- Implementation of cosine similarity for detecting phishing patterns
- A two-layer phishing detection system combining URL and HTML analysis
- A classification module that categorizes websites as safe, suspicious, or phishing
- A warning system that alerts users when a phishing website is detected
- A dataset of phishing and legitimate website samples used for testing
- Performance evaluation results including accuracy and response time
- Complete project documentation including report and source code

The final system is lightweight, privacy-preserving, and suitable for real-time deployment in browser environments. It provides an effective solution for protecting users against phishing attacks.

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